



B.Tech. CSE VI CSS 338
EMBEDDED SYSTEMS LAB

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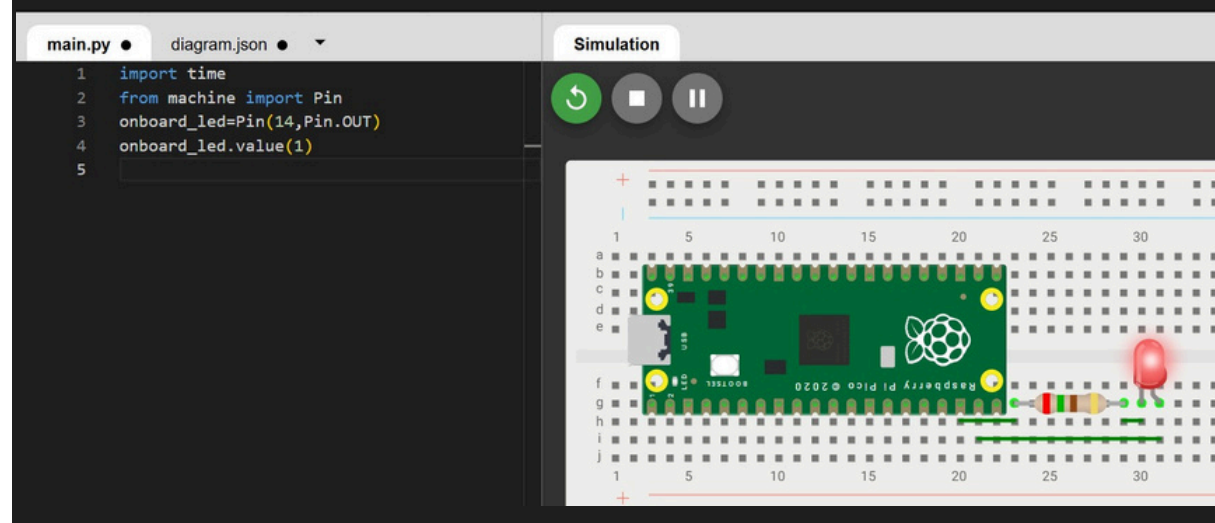
Name:Samina Parveen
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Embedded Systems Lab-Assignment1

Using **MicroPython** Simulating the following on a **Raspberry Pi Pico**:

1.Turn on a Led:

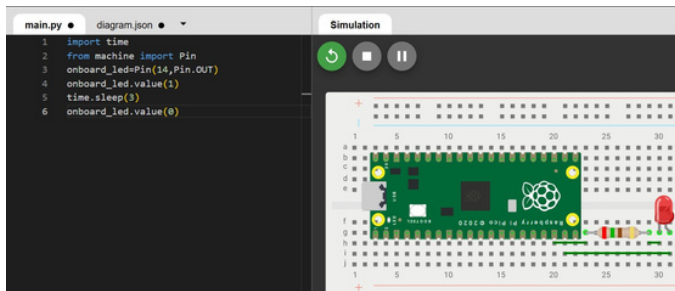
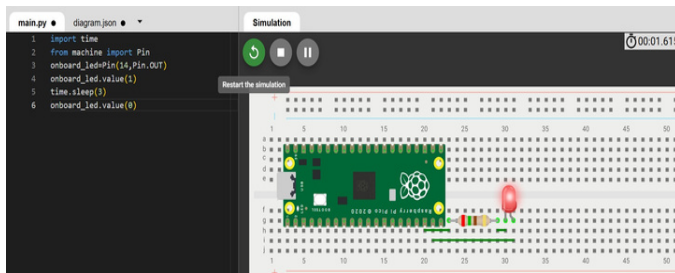
Code:

```
import time
from machine import Pin
onboard_led=Pin(14,Pin.OUT)
onboard_led.value(1)
```



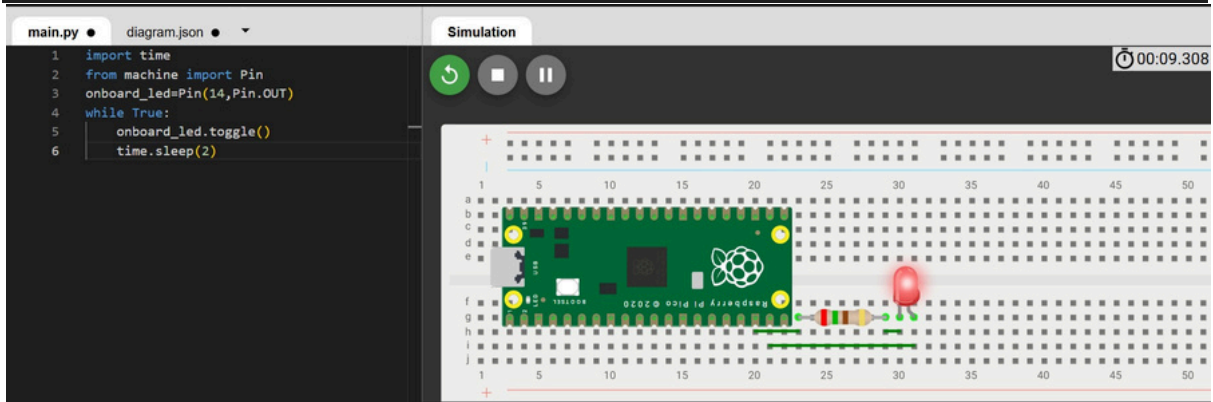
2.Turn on a Led and turn it off after 3 seconds

```
import time
from machine import Pin
onboard_led=Pin(14,Pin.OUT)
onboard_led.value(1)
time.sleep(3)
onboard_led.value(0)
```



3. Continuously Toggle a Led at an interval of 2 seconds

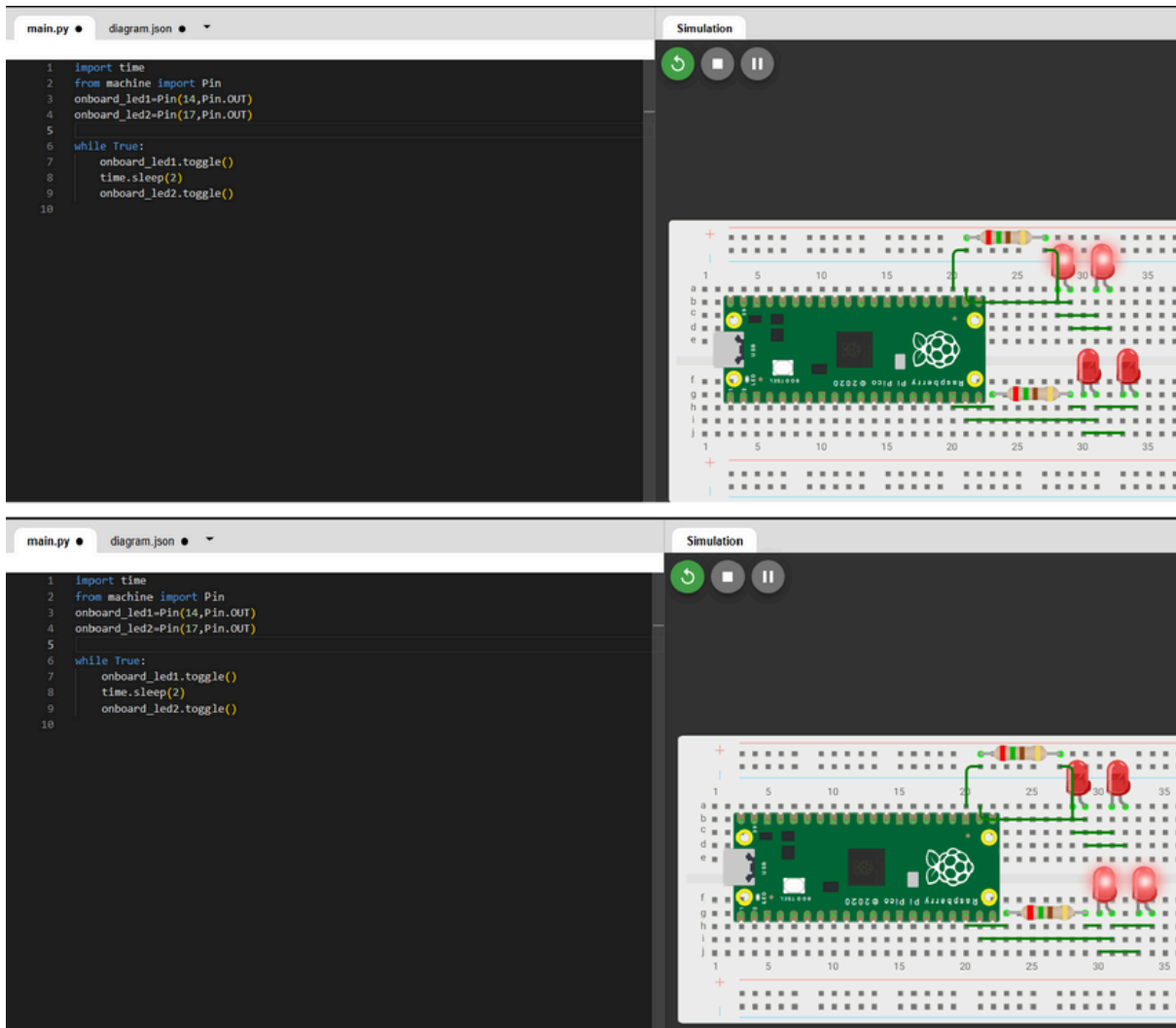
```
import time
from machine import Pin
onboard_led=Pin(14,Pin.OUT)
while True:
    onboard_led.toggle()
    time.sleep(2)
```



4. Continuously toggle two pairs of LEDs alternately every 2 seconds:

```
import time from machine
import Pin
onboard_led1=Pin(14,Pin.OUT)
onboard_led2=Pin(17,Pin.OUT)

while True:
    onboard_led1.toggle()
    time.sleep(2)
    onboard_led2.toggle()
```



5) Simulate a traffic light system using MicroPython on a Raspberry Pi Pico. The system should turn the lights Red, Orange, and Green in sequence. Each light must remain ON for a random duration measured in seconds, where the waiting time is never less than 2 seconds and never more than 5 seconds before switching to the next light.

```
import time

from machine import Pin

import random

onboard_led1=Pin(17,Pin.OUT)

onboard_led2=Pin(18,Pin.OUT)

onboard_led3=Pin(14,Pin.OUT)

while True:

    onboard_led3.value(0)

    onboard_led1.value(1)
```

```

time.sleep(random.randint(2,5))

onboard_led1.value(0)

onboard_led2.value(1)

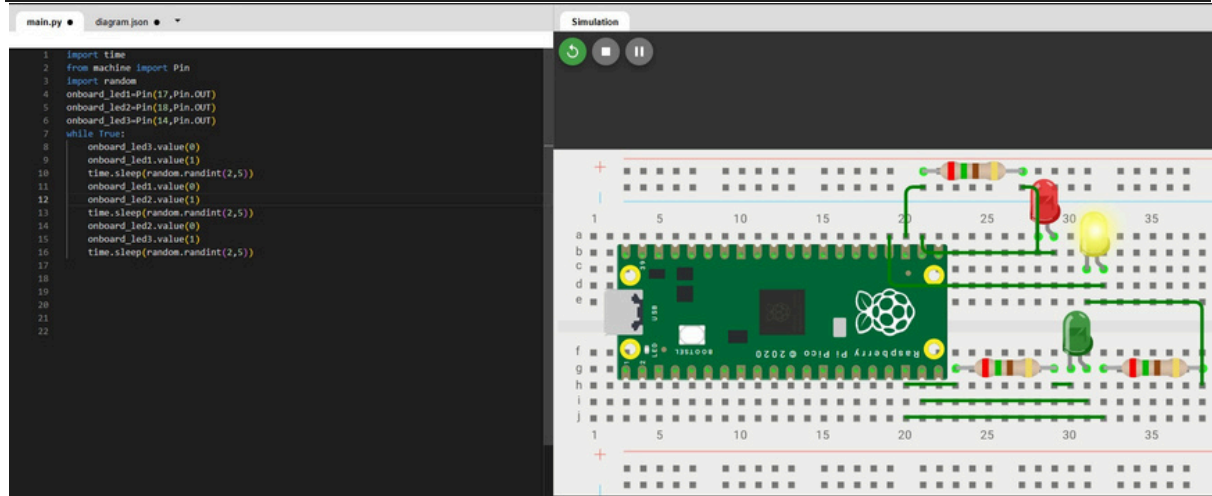
    time.sleep(random.randint(2,5))

onboard_led2.value(0)

onboard_led3.value(1)

time.sleep(random.randint(2,5))

```



6. Simulate a two-input truth table using three LEDs on a Raspberry Pi Pico using MicroPython.

```

import time

from machine import Pin

import random

onboard_led1=Pin(17,Pin.OUT)

onboard_led2=Pin(18,Pin.OUT)

OUT=Pin(14,Pin.OUT)

cases = [(0,0),(0,1),(1,0),(1,1)]

while True:

```

```

for a,b in cases:

    onboard_led1.value(a)

    onboard_led2.value(b)

    OUT.value(a and b)

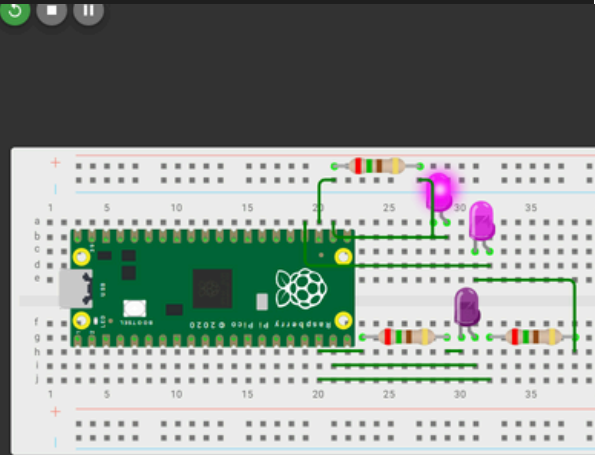
    time.sleep(2)

```

```

1 import time
2 from machine import Pin
3 import random
4 onboard_led1=Pin(17,Pin.OUT)
5 onboard_led2=Pin(18,Pin.OUT)
6 OUT=Pin(14,Pin.OUT)
7 cases = [(0,0),(0,1),(1,0),(1,1)]
8
9 while True:
10     for a,b in cases:
11         onboard_led1.value(a)
12         onboard_led2.value(b)
13         OUT.value(a and b)
14         time.sleep(2)
15
16
17

```

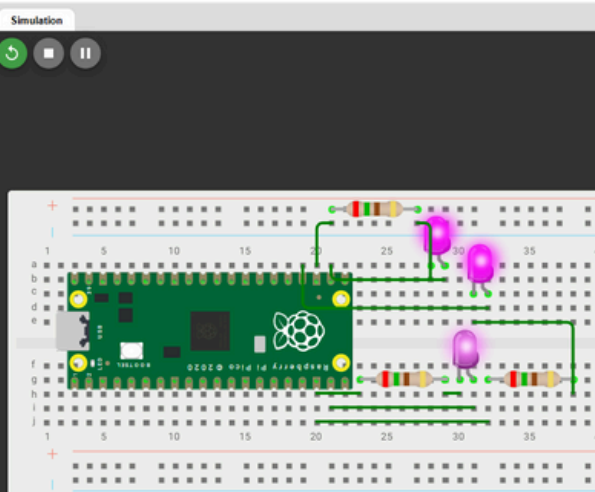


main.py • diagram.json •

```

1 import time
2 from machine import Pin
3 import random
4 onboard_led1=Pin(17,Pin.OUT)
5 onboard_led2=Pin(18,Pin.OUT)
6 OUT=Pin(14,Pin.OUT)
7 cases = [(0,0),(0,1),(1,0),(1,1)]
8
9 while True:
10     for a,b in cases:
11         onboard_led1.value(a)
12         onboard_led2.value(b)
13         OUT.value(a and b)
14         time.sleep(2)
15
16
17

```



7. Create a system of eight LEDs arranged in a circular fashion using a Raspberry Pi 1 and MicroPython. Create a simulation by randomly turning on the LEDs in sequence. After completion of a complete round decrease the time between the LEDs turning on by an amount delta.

```
1  from machine import Pin
2  import time
3
4  led_pins = [0, 1, 2, 3, 4, 5, 6, 7]
5  leds = [Pin(pin, Pin.OUT) for pin in led_pins]
6
7  current_delay = 0.5
8  delta = 0.05
9  min_delay = 0.05
10
11 while True:
12     for led in leds:
13         led.value(1)
14         time.sleep(current_delay)
15         led.value(0)
16
17     if current_delay > min_delay:
18         current_delay -= delta
19         current_delay = round(current_delay, 2)
20
21     if current_delay < min_delay:
22         current_delay = min_delay
23
```

